Spatial rescue in a biofilm under antibiotic treatment: a stochastic chain model

Abstract

Biofilms are spatially structured communities of bacteria in which cells near the surface are exposed to antibiotic while interior cells are shielded. We study how this structure affects the probability that a resistant mutant arises and rescues the population before it goes extinct under lethal antibiotic doses. We formulate a stochastic 1D "chain model" in which births and deaths drive a conveyor-belt flow of cells between a protected core and an exposed edge. We derive analytical approximations for the probability of mutant appearance as a function of the initial population size, edge width, and excess death rate, then validate and extend these results with stochastic simulations. **[TODO: Fill in main quantitative results once sweeps are complete.]**

1 Introduction

Bacteria growing in biofilms are notoriously difficult to eradicate with antibiotics [**?**]. One contributing factor is spatial structure: cells embedded in a biofilm matrix are shielded from diffusing drug, so only the outermost cells experience lethal concentrations. This creates an implicit refuge that could, in principle, extend the window during which a resistant mutant can arise and establish.

We ask a precise version of this question. Suppose a biofilm is exposed to an antibiotic at a dose above the minimum inhibitory concentration (MIC), so that surface cells die faster than they can replicate. The biofilm will eventually collapse. Can the spatial structure give the population more time—and more births—to produce a resistant mutant that rescues the lineage?

We compare two scenarios:

1. **Well-mixed:** All N_0 cells experience the full antibiotic dose simultaneously.
2. **Biofilm (chain model):** Only the outermost l cells (the "edge") are exposed; the remaining $N_0 - l$ cells occupy a protected "core."

The comparison is made under identical total population sizes and mutation rates. Any difference in rescue probability is purely a consequence of spatial structure.

2 Model

2.1 Population structure

The biofilm is represented as a 1D array of $N(t)$ cells, indexed from 0 (deep interior) to $N(t)-1$ (outermost surface). Two compartments are defined by the parameter l (edge width, in cells):

$$\text{Core} = \text{cells } 0, 1, \ldots, b-1, \quad b \equiv \max(0,\ N-l), \tag{1}$$

$$\text{Edge} = \text{cells } b, b+1, \ldots, N-1. \tag{2}$$

When $N \leq l$ the core is empty and all cells are in the edge. The edge always contains exactly $\min(N, l)$ cells; the core contains $\max(0, N-l)$.

2.2 Birth–death dynamics (Gillespie algorithm)

Events are drawn via the standard Gillespie algorithm (next-reaction method). Each event type fires at a total rate equal to its per-capita rate times the number of eligible cells. The per-capita rates are:

Compartment	Genotype	Birth rate	Death rate	Notes
Edge	WT	$b_e = 1$	$d_e = \text{dose}$	sweep variable
Edge	R	$b_R = 1$	$d_R = 0.6$	net positive growth
Core	WT	b_c	d_c	$b_c = d_c$ at steady state
Core	R	b_c	d_c	same as core WT (neutral refuge)

Time is measured in edge generations ($b_e \equiv 1$). The default parameter values used in simulations are given in Table 1.

Birth mechanics. When any cell in compartment c gives birth, a daughter is inserted at a uniformly random position within c. Because the array shifts to accommodate the new cell, the boundary $b = \max(0, N-l)$ increases by one: every birth in the edge pushes one edge cell into the core, and every birth in the core simply enlarges the core.

Death mechanics. When any cell dies it is removed from the array. The boundary b decreases by one: every death in the edge exposes one core cell to the edge, and every death in the core simply shrinks the core.

Conveyor-belt coupling. The net effect is that cells flow inward during edge births and outward during edge deaths. A core-born resistant mutant can therefore reach the edge and eventually dominate it—the *core-delivery* rescue pathway—even without any directed migration.

2.3 Mutation

Resistance is modeled as an irreversible point mutation. With probability μ per replication event, a daughter cell is assigned a new resistant (R) lineage rather than inheriting the parent's genotype. Back-mutation is ignored. Each de-novo lineage is tracked independently to determine its origin compartment, time of appearance, and eventual fate.

2.4 Rescue condition and simulation termination

Treatment begins at $t = 0$ with the edge WT death rate jumping to d_e = dose. Pre-treatment, $d_e = b_e = 1$ so the population is at steady state. A simulation ends when one of the following occurs:

- **Rescue:** the number of R cells in the edge reaches the threshold $r^* = 10$.
- **Extinction:** $N = 0$.
- **Timeout:** a maximum number of events or elapsed time is reached.

Table 1: Default simulation parameters.

Parameter	Value	Description
N_0	1000	Initial population size
l	100	Edge width (biofilm); $l = 2000$ for well-mixed control
b_e	1.0	WT edge birth rate (sets time unit)
$b_c = d_c$	0.2	Core birth/death rates (neutral core)
b_R	1.0	R edge birth rate
d_R	0.6	R edge death rate ($s_R = 0.4$)
μ	10^{-4}	Mutation probability per birth (main sweep)
r^*	10	Rescue threshold (R edge cells)
d_e	1.05–2.80	Sweep variable (log-spaced in $s = d_e - 1$)

3 Analytical approximations

We derive approximate closed-form expressions for the probability that at least one resistant mutant arises before the wild-type population goes extinct. This is a necessary—but not sufficient—condition for rescue; the additional requirements for establishment are analyzed in Section 3.3 and captured fully by the stochastic simulation.

3.1 Well-mixed population

Consider first a population of N_0 cells all exposed to the same antibiotic dose, with per-capita WT birth and death rates b_e and d_e. Define the net per-capita death rate:

$$s_e \equiv d_e - b_e = \text{dose} - 1 > 0. \tag{3}$$

In the deterministic limit the population declines exponentially:

$$N(t) = N_0\, e^{-s_e t}, \tag{4}$$

reaching effective extinction ($N = 1$) at time $T = \ln N_0 / s_e$.

Because mutations are replication-linked (probability μ per birth), the total mutation rate at time t is

$$\lambda(t) = \mu\, b_e\, N(t). \tag{5}$$

Mutations arrive as an inhomogeneous Poisson process, so the probability of *no* mutant arising by time T is

$$P(\text{no mutant}) = \exp\left(-\int_0^T \lambda(t)\, dt\right). \tag{6}$$

Evaluating the integral:

$$\int_0^T \mu\, b_e\, N_0\, e^{-s_e t}\, dt = \frac{\mu\, b_e\, N_0}{s_e}\left(1 - \frac{1}{N_0}\right) \approx \frac{\mu\, b_e\, N_0}{s_e}, \tag{7}$$

where the approximation holds for $N_0 \gg 1$. Therefore:

$$\boxed{P_{\text{WM}}(\text{mutant before extinction}) = 1 - \exp\left(-\frac{\mu\, b_e\, N_0}{s_e}\right).} \tag{8}$$

The dimensionless group $\mu b_e N_0 / s_e$ is the expected total number of mutations produced over the entire decline. When it is much less than one, $P_{\text{WM}} \approx \mu b_e N_0 / s_e$; when much greater than one, rescue is nearly certain.

3.2 Biofilm: two-compartment, two-phase analysis

In the biofilm, the population dynamics differ from the well-mixed case because core and edge cells have different rates and because the edge size is fixed at l cells as long as $N > l$.

Phase 1: core and edge coexist ($N > l$)

While $N > l$, the core contains $N - l$ cells and the edge contains l cells. The expected rate of change of total population size is:

$$\frac{dN}{dt} = (b_c - d_c)(N - l) + (b_e - d_e)\,l = -s_c(N - l) - s_e\,l, \tag{9}$$

where $s_c \equiv d_c - b_c$. With the default parameters $b_c = d_c = 0.2$, so $s_c = 0$, and (9) simplifies to:

$$\frac{dN}{dt} = -s_e\,l \quad (s_c = 0), \tag{10}$$

a *linear* decline at a constant rate $v \equiv s_e\,l$ cells per unit time. Intuitively, only the l edge cells are dying faster than they reproduce; the core maintains itself exactly and acts as a reservoir that continuously replenishes the edge. The population reaches $N = l$ at time:

$$T_1 = \frac{N_0 - l}{s_e\,l}. \tag{11}$$

The total mutation rate during Phase 1 is:

$$\lambda_1(t) = \mu\,[b_c(N(t) - l) + b_e\,l] = \mu\,[b_c(N_0 - l - s_e\,l\,t) + b_e\,l]. \tag{12}$$

Integrating over Phase 1:

$$\begin{aligned}\int_0^{T_1} \lambda_1(t)\,dt &= \mu\left[b_c\frac{(N_0 - l)^2}{2\,s_e\,l} + b_e\,l\,T_1\right] \\ &= \mu\left[\frac{b_c(N_0 - l)^2}{2\,s_e\,l} + \frac{b_e(N_0 - l)}{s_e}\right].\end{aligned} \tag{13}$$

Phase 2: edge only ($N \leq l$)

Once the core is exhausted the population follows exponential decline as in the well-mixed case, but starting from $N = l$:

$$N(t) = l\,e^{-s_e(t - T_1)}, \quad t \geq T_1. \tag{14}$$

The mutation contribution from Phase 2 is, by direct analogy with (8):

$$\int_{T_1}^{T_2} \lambda_2(t)\,dt \approx \frac{\mu\,b_e\,l}{s_e}. \tag{15}$$

Combined probability

Defining the total expected mutation count $\Lambda \equiv \Lambda_1 + \Lambda_2$ (the sum of (13) and (15)):

$$\boxed{P_{\mathrm{BF}}(\text{mutant before extinction}) = 1 - e^{-\Lambda},} \tag{16}$$

$$\Lambda = \mu\left[\frac{b_c(N_0 - l)^2}{2\,s_e\,l} + \frac{b_e(N_0 - l)}{s_e} + \frac{b_e\,l}{s_e}\right] = \frac{\mu}{s_e}\left[\frac{b_c(N_0 - l)^2}{2\,l} + b_e\,N_0\right]. \tag{17}$$

Comparison to well-mixed: effect of core birth rate on mutation supply

The ratio $\Lambda/\Lambda_{\mathrm{WM}}$ quantifies the biofilm's mutation-supply advantage:

$$\frac{\Lambda}{\Lambda_{\mathrm{WM}}} = 1 + \frac{b_c(N_0 - l)^2}{2\, b_e\, N_0\, l}. \tag{18}$$

This ratio is always ≥ 1. For the default parameters ($N_0 = 1000$, $l = 100$, $b_c = b_e = 1$) it equals $1 + 4.05 = 5.05$: the biofilm expects five times as many mutations as the well-mixed control.

Dormant-core limit ($b_c \to 0$). Setting $b_c = 0$ in equation (17):

$$\Lambda\Big|_{b_c=0} = \frac{\mu\, b_e\, (N_0 - l)}{s_e} + \frac{\mu\, b_e\, l}{s_e} = \frac{\mu\, b_e\, N_0}{s_e} = \Lambda_{\mathrm{WM}}. \tag{19}$$

A completely dormant core ($b_c = d_c = 0$) provides *exactly as many mutation opportunities as the well-mixed control.* The core extends the population's lifetime—core cells are not killed by the antibiotic during Phase 1—but without births it contributes zero mutations. The entire mutation-supply advantage of the biofilm therefore comes from core births alone; the surplus in equation (18) is exactly zero when $b_c = 0$ and grows linearly with b_c thereafter.

3.3 Two effects of core metabolic activity on the core-delivery pathway

Core metabolic activity—the turnover rate $\rho \equiv b_c = d_c$ at which core cells divide and die—drives a fundamental tension in the core-delivery rescue pathway. The mutation-supply analysis above captures only one side of it:

Effect 1 (mutation supply, captured by Λ). Every core birth is a mutation opportunity. The core-delivery term in Λ grows linearly with b_c (equation (18)), and vanishes in the dormant limit (equation (19)).

Effect 2 (genetic drift, not captured by Λ). A resistant lineage born in the core lives in a neutral environment ($b_{R,c} = b_c$, $d_{R,c} = d_c$): it has no growth advantage there. It must survive *neutral genetic drift* long enough to be conveyed to the edge by the retreating boundary. Higher turnover means faster drift, so most lineages are lost before delivery.

We quantify Effect 2 and show how the two combine.

Lineage survival under neutral drift in the core

A mutant born at depth d from the core–edge boundary (i.e. d positions away from being edge-exposed) must wait for the boundary to retreat past it. Since the boundary retreats at rate $s_e\, l$ cells per unit time (equation (10)), the expected waiting time is

$$\tau_d \approx \frac{d}{s_e\, l}. \tag{20}$$

During this wait, the lineage undergoes a critical ($b_c = d_c$) birth-death process. Starting from a single cell, the probability of surviving to time τ is [?]

$$P(\text{survive to delivery} \mid d) = \frac{1}{1 + b_c\, \tau_d} = \frac{1}{1 + b_c\, d \,/\, (s_e\, l)}. \tag{21}$$

Since births are placed at uniformly random depths $d \in [0,\, N_0 - l]$ within the core, the mean survival probability over all birth positions is

$$\langle P(\text{survive to delivery}) \rangle = \frac{\ln(1+\alpha)}{\alpha}, \qquad \alpha \equiv \frac{b_c\,(N_0 - l)}{s_e\, l}, \tag{22}$$

where α is the dimensionless ratio of core turnover to boundary retreat rate. In the two limits:

$$\langle P \rangle \;\to\; 1 \qquad\qquad \alpha \to 0 \;\; \text{(dormant: every lineage survives until delivery)}, \tag{23}$$

$$\langle P \rangle \;\to\; \frac{\ln \alpha}{\alpha} \;\to\; 0 \quad \alpha \to \infty \;\; \text{(very active: most lineages lost to drift)}. \tag{24}$$

Net rate of core lineages reaching the edge

Multiplying the total core mutation rate by the mean survival probability gives the rate at which core mutant lineages are successfully delivered to the edge:

$$\begin{aligned} \Gamma_{\text{core}} &= \underbrace{\mu\, b_c\,(N_0 - l)}_{\text{Effect 1: mutation rate}} \times \underbrace{\frac{\ln(1+\alpha)}{\alpha}}_{\text{Effect 2: mean survival}} \\ &= \mu\, s_e\, l\, \ln\!\left(1 + \frac{b_c(N_0 - l)}{s_e\, l}\right). \end{aligned} \tag{25}$$

In the two limits:

$$\Gamma_{\text{core}} \approx \mu\, b_c\,(N_0 - l) \qquad\qquad \alpha \to 0 \;\; \text{(linear in } b_c\text{; drift negligible)}, \tag{26}$$

$$\Gamma_{\text{core}} \approx \mu\, s_e\, l\, \ln\!\left(\frac{b_c(N_0 - l)}{s_e\, l}\right) \qquad \alpha \to \infty \;\; \text{(logarithmic saturation)}. \tag{27}$$

At high turnover, increasing b_c yields only logarithmic gains in delivered lineages: the additional mutations from faster turnover are nearly all eliminated by drift before reaching the edge. Effects 1 and 2 nearly cancel.

Bolus delivery and the rescue threshold

A lineage that survives to delivery does not arrive at the edge as a single cell. For a critical birth-death process conditioned on survival to time τ, the expected lineage size at delivery is [?]

$$\mathbb{E}[\,\text{size} \mid \text{survive } \tau\,] = 1 + b_c\,\tau. \tag{28}$$

Surviving lineages from an active core therefore arrive as a multi-cell *bolus*. The martingale property of critical birth-death processes—the expected lineage size including extinct lineages (counted as zero) is preserved at all times—implies that Effects 2 and 3 cancel exactly *in expectation*:

$$P(\text{survive}) \;\times\; \mathbb{E}[\,\text{size} \mid \text{survive}\,] \;=\; \frac{1}{1+b_c\tau} \times (1+b_c\tau) \;=\; 1 \quad \text{for all } b_c \text{ and } \tau. \tag{29}$$

If rescue probability were linear in the number of cells delivered, turnover would have no net effect on the core-delivery pathway. The rescue threshold $r^* = 10$ is a discrete, nonlinear criterion: k cells arriving simultaneously are far more likely to cross it than k single-cell arrivals separated in time. The martingale cancellation therefore does not hold at the level of rescue events, and an active core retains a residual advantage through bolus delivery that Λ does not capture.

3.4 The dose-dependent transition: nursery vs. drift trap

A direct consequence of the dormant-core limit (equation (19)) reframes the entire dose-dependence. The passive buffer role of the core—extending the population's lifetime—provides *no rescue advantage* over the well-mixed case: the longer Phase 1 duration exactly compensates for having fewer cells in the edge at any given time, leaving $\Lambda_{\text{edge}} = \Lambda_{\text{WM}}$ regardless of dose. It follows that

> *The entire rescue advantage of the biofilm over the well-mixed case comes from core births.*

Whether those births translate into rescue depends on α. The result is a clean two-regime picture:

	Low selection ($\alpha \gg 1$)	High selection ($\alpha \ll 1$)
Dormant core ($b_c = 0$)	$P_{\text{BF}} \approx P_{\text{WM}}$	$P_{\text{BF}} \approx P_{\text{WM}}$
Active core ($b_c > 0$)	$P_{\text{BF}} \approx P_{\text{WM}}$	$P_{\text{BF}} \gg P_{\text{WM}}$

At low selection the active core is no better than a dormant one: core mutations arise but are lost to drift before delivery ($\langle P(\text{survive})\rangle \to 0$), and rescue probability collapses toward the well-mixed value. At high selection the active core is decisive: the boundary retreats fast enough that drift has little time to act, and the core functions as an efficient nursery. A dormant core, by contrast, contributes nothing to rescue at any dose. *Note: this table captures drift effects on the core-delivery pathway only. A second suppression mechanism—WT resupply into the edge—is analyzed separately in Section 3.5 and can drive $P_{\mathrm{BF}} < P_{\mathrm{WM}}$ at low selection even in the dormant-core case.*

The irony. The same antibiotic selection pressure that endangers the population is the mechanism that makes the core valuable. High s_e simultaneously accelerates the collapse of the wild-type edge *and* drives the boundary retreat that delivers core-born mutants before drift can eliminate them. The biofilm converts the killing force of the antibiotic into a delivery mechanism for resistance.

The crossover. The transition between regimes occurs at $\alpha = 1$:

$$s_e^* = \frac{b_c(N_0 - l)}{l}. \tag{30}$$

For default parameters, $s_e^* = 0.2 \times 900/100 = 1.8$ (dose ≈ 2.8). The entire swept dose range $s_e \in [0.05, 2.0]$ therefore sits at or below the crossover, placing most simulated conditions in the drift-suppressed regime and making the upper end of the sweep the most mechanistically interesting region.

Prediction for the b_c sensitivity sweep. Since the biofilm rescue advantage depends on b_c only through core-delivery, and core-delivery is drift-suppressed at low selection, the rescue curves for different values of b_c should *fan out* with increasing dose:

- At low dose: all b_c values give $P_{\text{rescue}} \approx P_{\mathrm{WM}}$ — curves collapse onto the well-mixed baseline.
- At high dose: P_{rescue} increases strongly with b_c — curves diverge from one another and from the well-mixed baseline.

The dose at which the curves begin to separate is set by equation (30), which predicts a different separation point for each b_c value. This gives a concrete, parameter-free test of the mechanism against the simulation sweep.

3.5 WT resupply and suppression of edge-born mutants

The mechanisms analyzed so far—mutation supply and core-delivery drift—concern core-born lineages. The conveyor-belt dynamics impose a second, distinct suppression mechanism on mutations arising directly in the edge, acting through a *wild-type resupply flux* from the core.

The mechanism

During Phase 1 the edge compartment always contains exactly l cells. Every edge death is immediately compensated by the outermost core cell moving into the edge. Because the core is wild-type dominated throughout Phase 1, each edge death injects a fresh WT cell into the edge. The total WT resupply rate is

$$m_{\mathrm{WT}} \approx d_e \cdot l \quad \text{(WT cells per unit time during Phase 1).} \tag{31}$$

In the well-mixed model there is no protected reservoir: dying WT cells are simply lost, and the WT pool shrinks monotonically. A resistant mutant in the WM population therefore faces progressively weaker competition as the WT decline. In the biofilm the WT pool in the edge is actively *maintained* by resupply from the core throughout Phase 1.

Effect on R establishment

Let k denote the number of R cells in the edge (held at l cells during Phase 1). The conveyor-belt mechanics—births push the innermost edge cell to the core, deaths draw the outermost core cell to the edge—yield effective transition rates for k:

$$\mathrm{Rate}(k \to k+1) = b_R\, k\, (l-k)/l, \tag{32}$$
$$\mathrm{Rate}(k \to k-1) = k\, [b_e\, (l-k)/l + d_R]\,. \tag{33}$$

The downward rate contains two contributions: WT births that push R cells into the core (rate $b_e(l-k)k/l$), and R deaths replenished by WT arriving from the core (rate $d_R k$). Both effects are sustained at a constant level throughout Phase 1 because the WT pool in the edge is maintained by resupply. This reduces the effective establishment probability $p_{\mathrm{est}}^{\mathrm{eff}}$ below the value $1 - d_R/b_R = 0.4$ that applies in a declining WM population.

An exact expression for $p_{\mathrm{est}}^{\mathrm{eff}}$ requires solving the first-passage problem for the Markov chain (32)–(33) with absorbing states $k = 0$ (extinction) and $k = r^* = 10$ (rescue), which does not simplify readily. The qualitative conclusion is that WT resupply suppresses edge-born R establishment, and this suppression is active for the full Phase 1 duration $T_1 = (N_0 - l)/v$.

Dose dependence

The suppression window has duration $T_1 = (N_0 - l)/v$, where $v = s_e\, l$ is the boundary retreat velocity. At low selection (v small, T_1 large), each edge-born R lineage faces sustained WT dilution before the core is exhausted; suppression is strong. At high selection (v large, T_1 small), Phase 2 dominates quickly and WT resupply has little time to act.

Importantly, WT resupply occurs even when $b_c = 0$ (dormant core), because core cells move to the edge upon edge deaths regardless of whether they divide. The dormant-core equality $\Lambda\big|_{b_c=0} = \Lambda_{\mathrm{WM}}$ is a statement about *mutation supply* only; WT resupply suppresses establishment probabilities through a channel not captured by Λ, and could cause $P^{\mathrm{BF}}_{\mathrm{rescue}} \leq P_{\mathrm{WM}}$ even in the dormant limit.

Combined suppression prediction

Adding WT resupply to the drift-suppression picture gives a *non-monotone* dose dependence for the biofilm rescue advantage:

	Low selection (v small, T_1 large)	High selection (v large, T_1 small)
Dormant core ($b_c = 0$)	$P_{\mathrm{BF}} \lesssim P_{\mathrm{WM}}$ (WT resupply)	$P_{\mathrm{BF}} \approx P_{\mathrm{WM}}$
Active core ($b_c > 0$)	$P_{\mathrm{BF}} < P_{\mathrm{WM}}$ (drift + resupply)	$P_{\mathrm{BF}} \gg P_{\mathrm{WM}}$

The crossover dose at which the biofilm transitions from suppression to enhancement is governed by the interplay of core drift (equation (30)) and WT resupply (equation (31)). Whether this crossover is empirically detectable, and at what dose it occurs, is a question for the stochastic simulation.

3.6 What the analytical approximation captures

Table 2 summarizes which effects are included in the approximation $P = 1 - e^{-\Lambda}$ and which are not.

Λ counts every mutation event produced, treating each as equally capable of rescue. The true rescue probability is therefore bounded above:

$$P(\text{rescue}) \;\leq\; 1 - e^{-\Lambda}. \tag{34}$$

The gap is set by establishment probabilities. For the *edge-delivery* pathway, a single R cell born in the edge has establishment probability $1 - d_R/b_R = 1 - 0.6 = 0.4$ (exact result for a supercritical birth-death process [**?**]), though this is attenuated by WT resupply throughout Phase 1 (Section 3.5). For the *core-delivery* pathway, establishment is further attenuated by the drift-survival factor $\ln(1+\alpha)/\alpha$ and modified nonlinearly by bolus delivery. Both pathways are tracked explicitly in the stochastic simulation (Section 4).

4 Simulation results

[TODO: This section will present results from the OSG parameter sweep: main rescue-vs-dose curves for all three conditions (biofilm, chainWM, wellMixed), pathway breakdown (edge-origin vs core-delivery rescue), and sensitivity analyses over μ and b_c.]

Table 2: Scope of the analytical approximation $P_{\mathrm{BF}} = 1 - e^{-\Lambda}$.

Effect	Direction	In Λ?
Core births increase mutation supply	$+$	✓
Core lifetime extends edge mutation window	$+$	✓
Drift eliminates core lineages before delivery	$-$	$\times$
Bolus delivery aids threshold crossing	$+$	$\times$
WT resupply suppresses edge-born establishment	$-$	$\times$
Establishment probability of edge-born mutants	—	$\times$

4.1 Rescue probability vs. dose

[TODO: Figure: P(rescue) vs. $s = \text{dose} - 1$ for the three conditions. Expected result: biofilm $>$ chainWM $\approx$ wellMixed at every dose, with largest gap at intermediate doses.]

4.2 Rescue pathway breakdown

[TODO: Figure: fraction of biofilm rescues attributed to edge-origin vs. core-delivery. Expected result: core-delivery fraction rises with dose, consistent with the core becoming more important as edge mutations become less likely to survive.]

4.3 Sensitivity to core rate (b_c)

[TODO: Figure: P(rescue) vs. dose for $b_c/b_e \in \{0.05, 0.1, 0.2\}$. Equation (18) predicts the rescue advantage scales as $b_c(N_0 - l)^2/(2lN_0b_e)$, so slower core growth should reduce the advantage.]

4.4 Sensitivity to mutation rate (μ)

[TODO: Figure: P(rescue) vs. dose for $\mu \in \{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$. In the rare-mutation regime ($\Lambda \ll 1$), P(rescue) $\propto \mu$, so the curves should shift vertically but preserve their shape.]

5 Discussion

[TODO: Discuss the mechanism: the core acts as a long-lived nursery for mutations that can be delivered to the edge by conveyor-belt flow, and as a buffer that slows the overall population decline and keeps the total birth rate high. Compare to Fruet et al. 2025 and other biofilm-rescue literature.]

A Derivation of the linear-decline approximation

Here we verify that the $s_c = 0$ case yields exactly linear decline. With $b_c = d_c$, the core birth and death rates cancel and equation (9) becomes:

$$\frac{dN}{dt} = -s_e\, l = \text{const}, \quad N > l. \tag{35}$$

Integrating from $t = 0$: $N(t) = N_0 - s_e\, l\, t$.

The total birth rate during Phase 1 is:

$$\begin{aligned} B(t) &= b_c(N(t) - l) + b_e\, l && (36)\\ &= b_c(N_0 - l - s_e\, l\, t) + b_e\, l. && (37) \end{aligned}$$

Integrating $\mu B(t)$ over $[0, T_1]$ with $T_1 = (N_0 - l)/(s_e l)$:

$$\begin{aligned} \Lambda_1 &= \mu \int_0^{T_1} B(t)\, dt && (38)\\ &= \mu \left[b_c \int_0^{T_1} (N_0 - l - s_e\, l\, t)\, dt + b_e\, l\, T_1 \right] && (39)\\ &= \mu \left[b_c \cdot \frac{(N_0 - l)}{2} \cdot T_1 + b_e\, l\, T_1 \right] && (40)\\ &= \mu\, T_1 \left[\frac{b_c(N_0 - l)}{2} + b_e\, l \right]. && (41) \end{aligned}$$

Substituting T_1 and combining with $\Lambda_2 = \mu b_e l / s_e$ recovers equation (17).